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Industrial Internship Report on

”URL Shortener using Python (Flask)”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT). This internship focused on developing a URL Shortener application using Python and Flask. The objective of the project was to convert long URLs into shorter, unique links and redirect users to the original URL when accessed. The project involved implementing backend logic using Flask, generating unique short codes, and storing URL mappings using an SQLite database. It also included handling user input through a simple web interface and implementing redirection functionality. This internship provided practical exposure to web development concepts, database integration, and application logic design. It helped in strengthening Python programming skills and understanding real-world application development. Overall, the internship was a valuable learning experience that enhanced both technical and problem-solving abilities.

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1 Preface

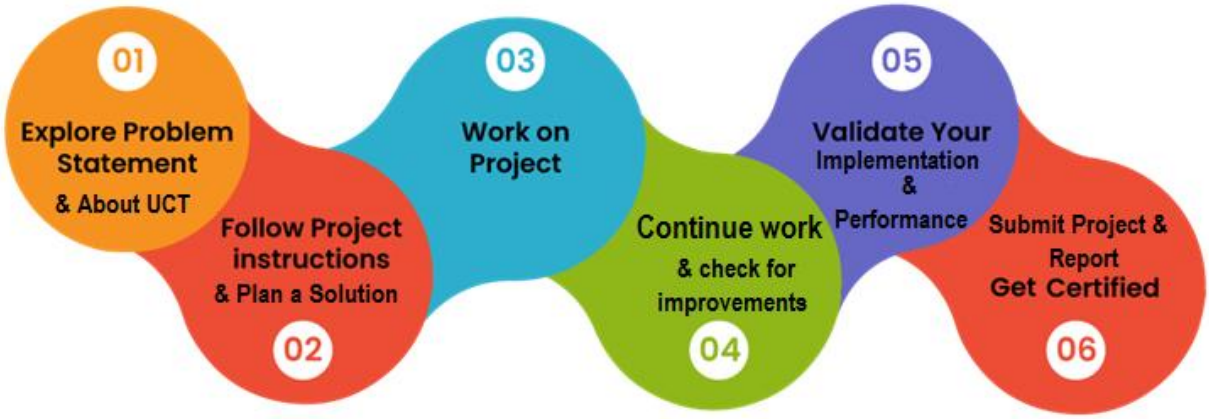
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

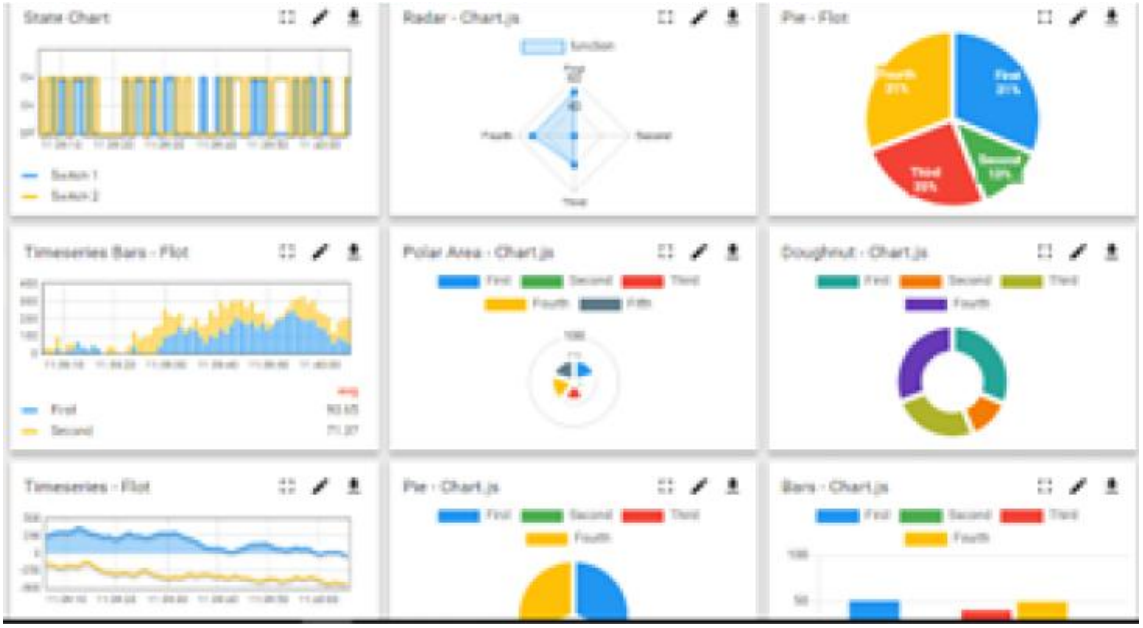
i. UCT IoT Platform (UCT Insight)

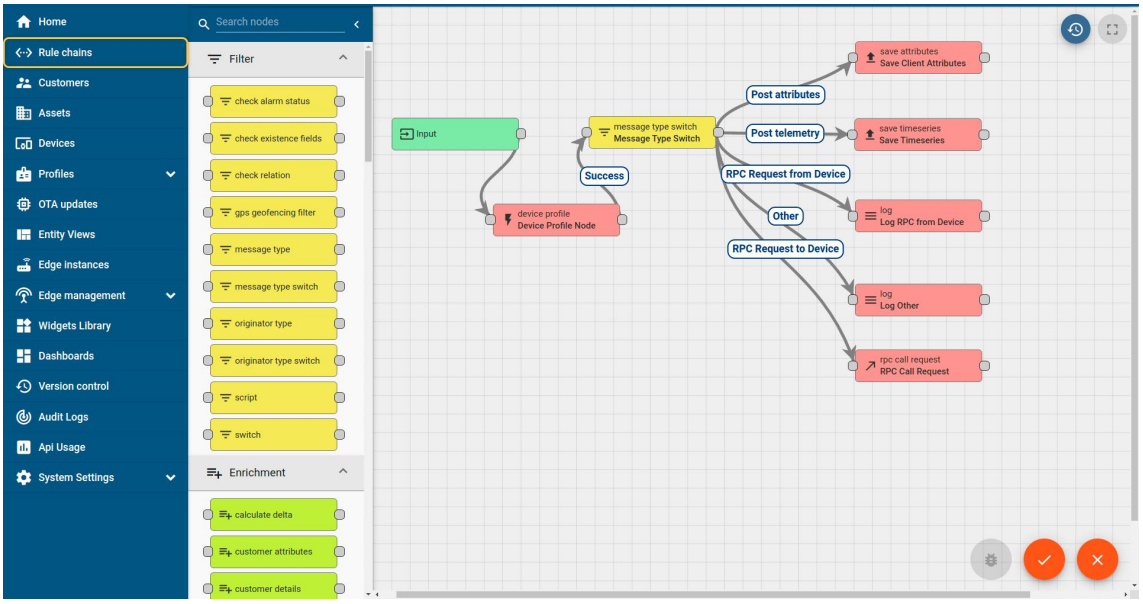
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine


State Chart
Radar - Chart.js
Pie - Flot
Timeseries Bars - Flot
Polar Area - Chart.js
Doughnut - Chart.js
Timeseries - Flot
Pie - Chart.js
Bars - Chart.js


Home
Rule chains
Customers
Assets
Devices
Profiles
OTA updates
Entity Views
Edge instances
Edge management
Widgets Library
Dashboards
Version control
Audit Logs
Api Usage
System Settings
Search nodes
Filter
check alarm status
check existence fields
check relation
gps geofencing filter
message type
message type switch
originator type
originator type switch
script
switch
Enrichment
calculate delta
customer attributes
customer details
Input
message type switch
Message Type Switch
device profile
Device Profile Node
Success
Post attributes
Post telemetry
RPC Request from Device
Other
RPC Request to Device
save attributes
Save Client Attributes
save timeseries
Save Timeseries
log
Log RPC from Device
log
Log Other
rpc call request
RPC Call Request

ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

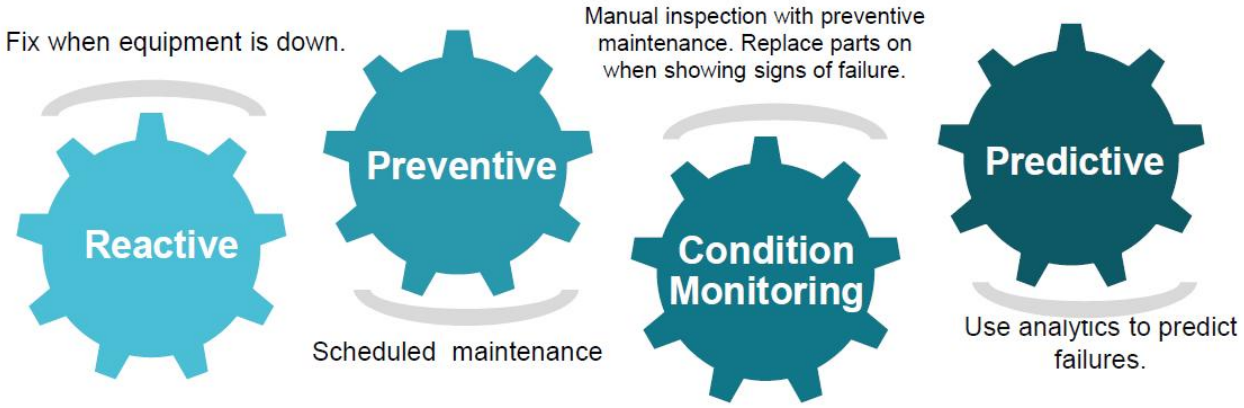
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

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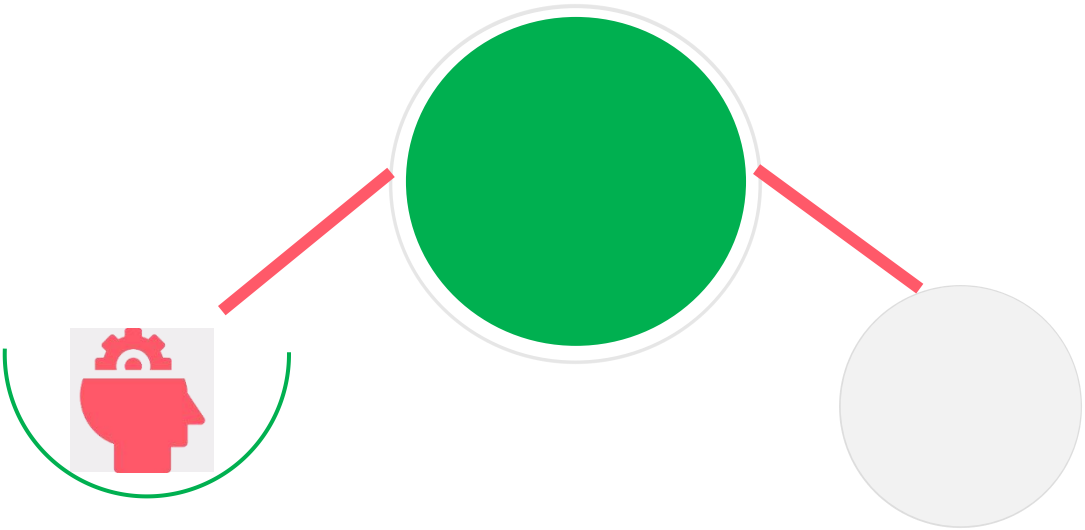
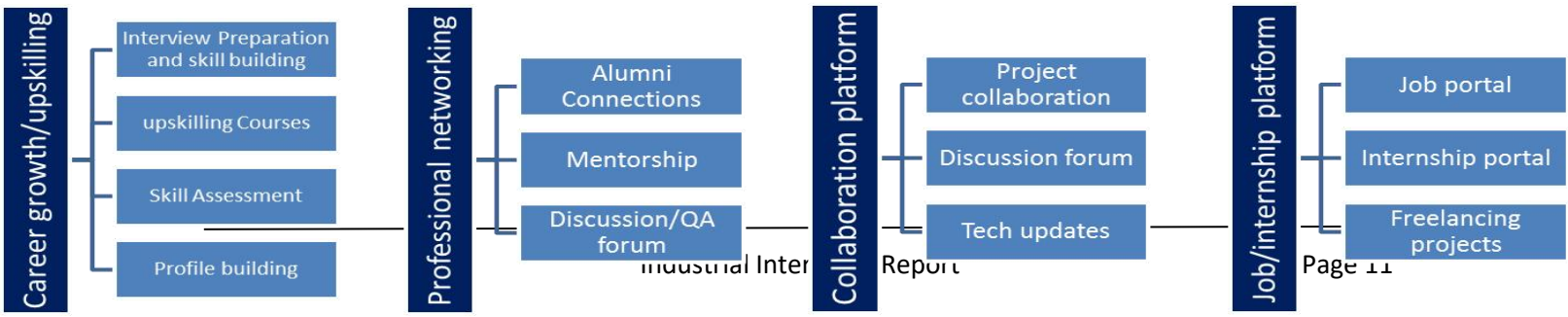


Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

[1]

[2]

[3]

2.6 Glossary

Terms	Acronym

3 Problem Statement

In the assigned problem statement

[The objective of this project is to develop a URL Shortener application using Python that converts long URLs into shorter and manageable links.

The system should:

- Accept a long URL as input from the user

- Generate a unique short code for the URL

- Store the mapping between original URL and short code

- Redirect users to the original URL when the short link is accessed

The goal is to create a simple, efficient, and user-friendly URL shortening system.]

4 Existing and Proposed solution

Existing URL shortener services like Bitly and TinyURL provide similar functionality but are web-based platforms with additional features and require internet access.

Limitations of existing solutions:

- Dependence on external services

- Limited control over data

- Not suitable for learning basic implementation

Proposed Solution:

The proposed solution is a simple URL Shortener built using Python and Flask, which runs locally and stores URL mappings in an SQLite database.

Value Addition:

- Easy to implement and understand

- Does not require external APIs

- Demonstrates backend development and database integration

- Suitable for learning real-world web application concepts

4.1 Code submission (Github link)

https://github.com/Venisha24/upskillcampus/blob/main/app.py

4.2 Report submission (Github link) :

https://github.com/Venisha24/upskillcampus/blob/main/URLShortener_Venisha_USC_UCT.pdf

5 Proposed Design/ Model

The system follows a simple web appcation architecture.

1. User enters a long URL in the web form.

2. Flask processes the request.

3. A unique short code is generated.

4. The mapping is stored in SQLite database.

5. A shortened URL is displayed to the user.

6. When the short URL is accessed, Flask retrieves the original URL from the database and redirects the user.

This flow ensures efficient URL shortening and redirection.

5.1 High Level Diagram (if applicable)

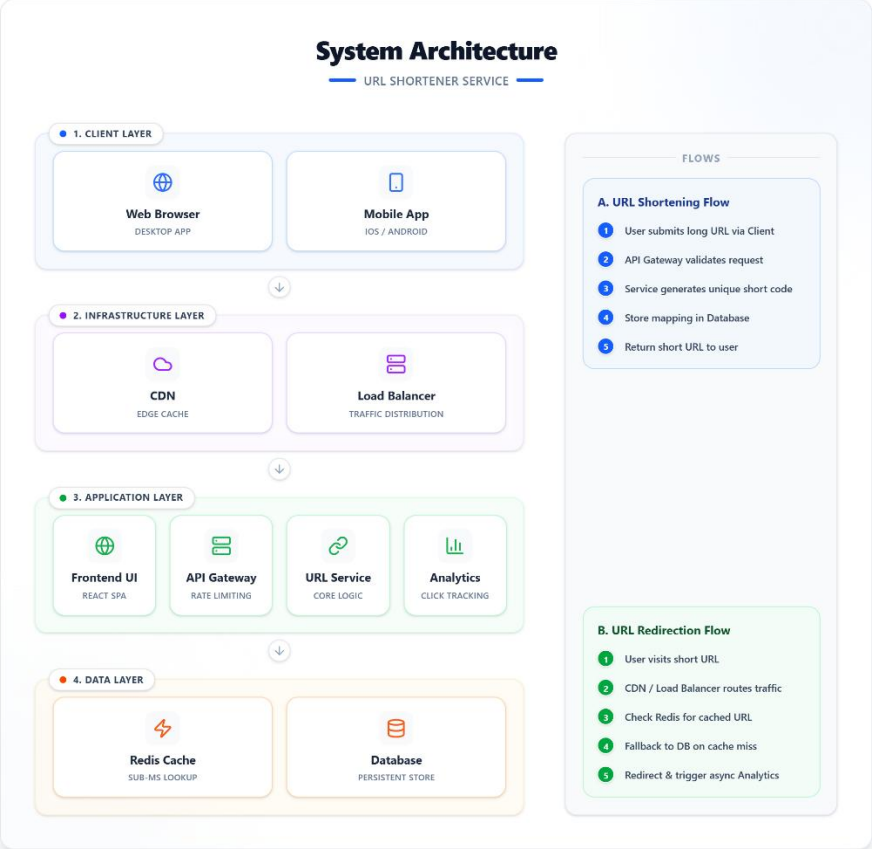


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

User → Flask App → Database (SQLite)

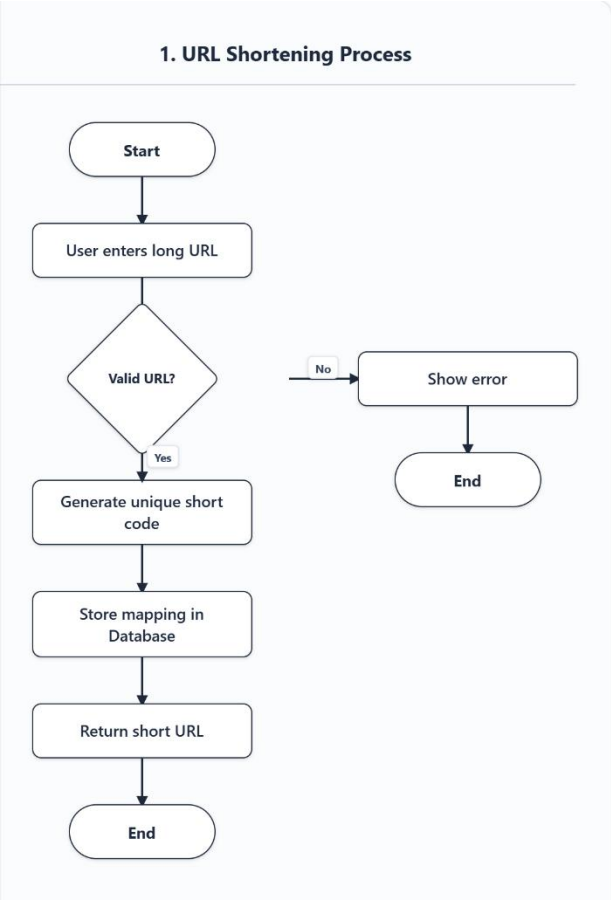
↓

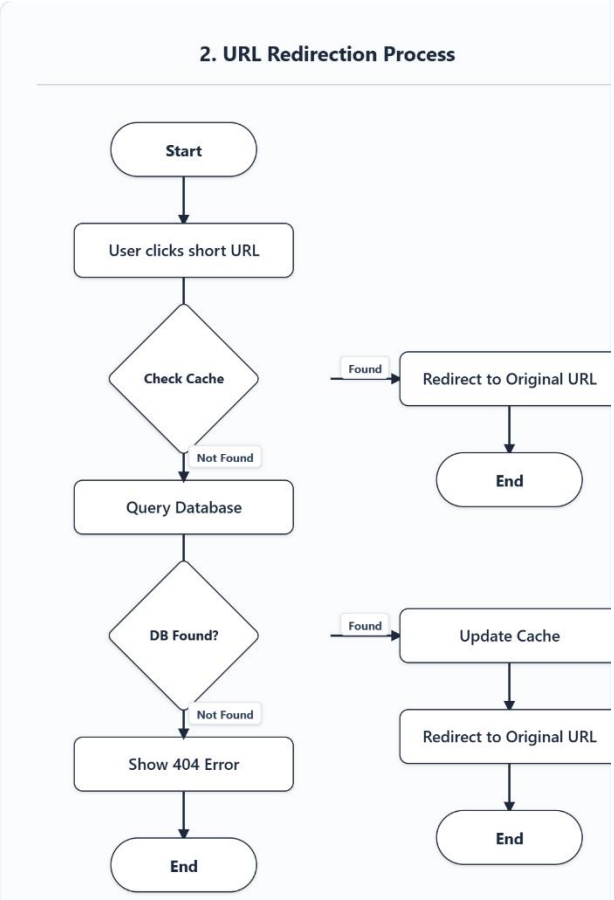
Short URL Generated

↓

Redirect to Original URL

5.3 Interfaces (if applicable)





6 Performance Test

6.1 Test Plan/ Test Cases

The system was tested for basic functionality and performance.

Constraints:
- Database storage capacity
- Response time for redirection

6.2 Test Procedure

Test Procedure:
- Multiple URLs were entered and shortened
- Short URLs were accessed to verify redirection

6.3 Performance Outcome

Performance Outcome:
- URLs were shortened successfully
- Redirection worked correctly
- System performed efficiently for small-scale usage

7 My learnings

During this project, I learned how to build a web application using Flask and integrate it with a database. I gained practical knowledge of handling user input, generating unique identifiers, and implementing redirection logic.

I also improved my understanding of backend development, routing, and database operations. This project helped strengthen my problem-solving skills and provided hands-on experience in real-world application development.

8 Future work scope

Future improvements can include:
- Adding user authentication
- Tracking number of clicks on shortened URLs
- Deploying the application online
- Improving UI with better frontend design
- Adding custom short URLs